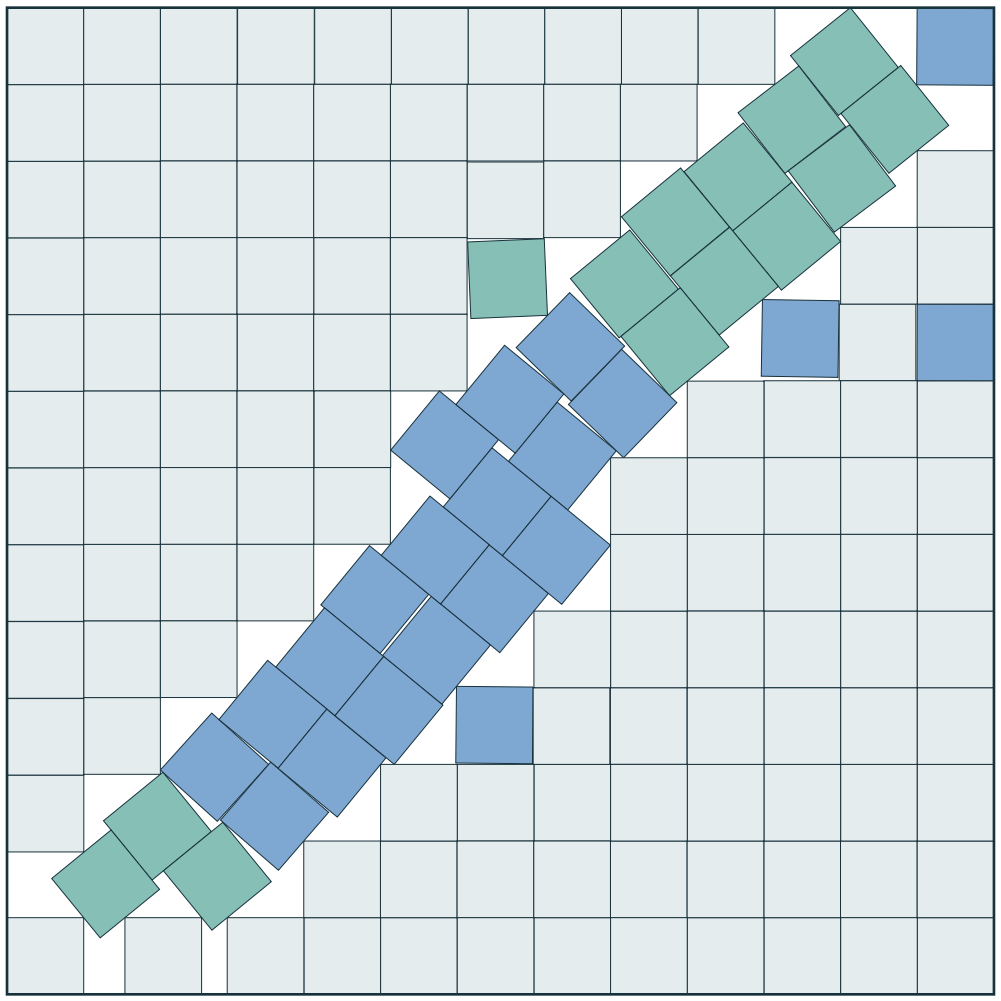
153 squares. A smaller enclosure.

A complete construction with a new contact return, compared with the current public benchmark.

PRIOR PUBLIC CONSTRUCTION

12.872029849082

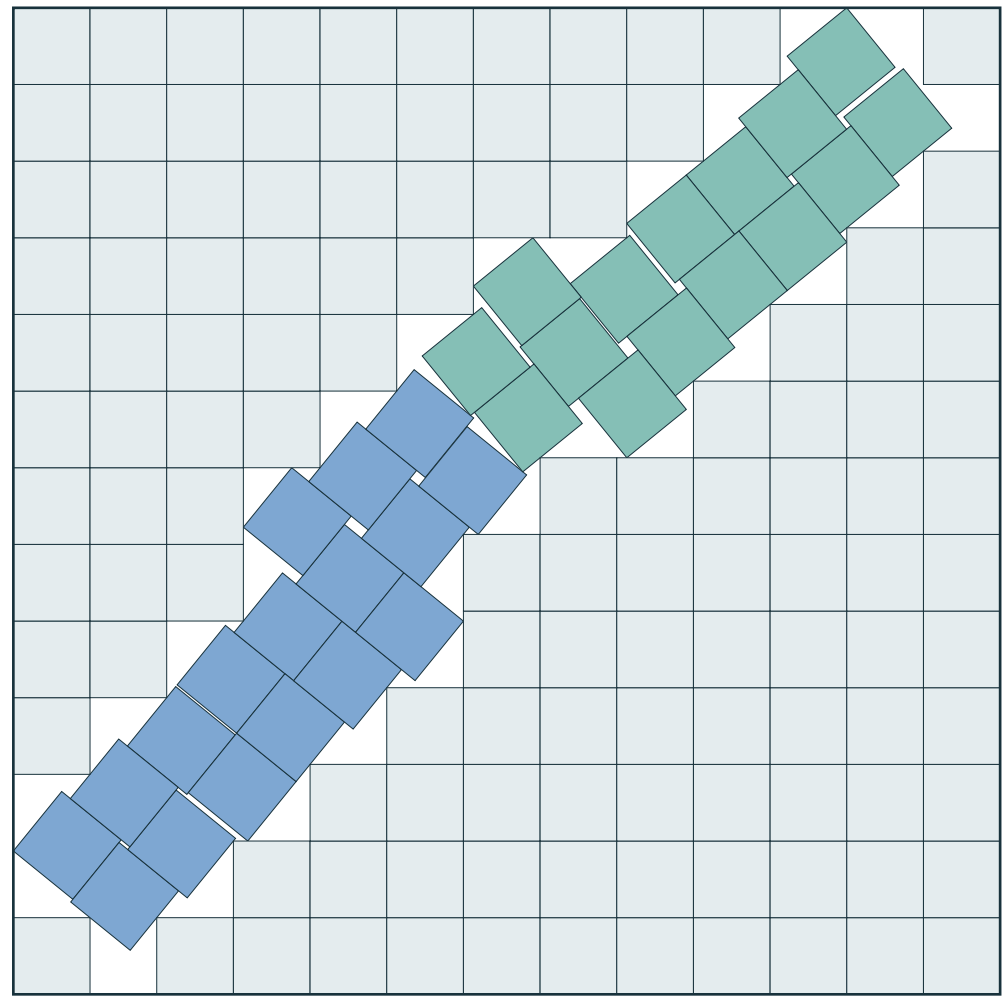
Nate Chaoweeraprasit / SQUISH



NEW VERIFIED CONSTRUCTION

12.870013850408

PingYou Ltd / both displayed sides rounded upward



Same scale: each small square has side one.

0.002015998674
reduction in enclosing side

0.051896
square units of area saved

0.01566%
smaller side, all 153 unit squares present

Benchmark checked 11 October 2026, 10:24 UTC: Square Packing Atlas and jlevy/squares issue #481. The diagrams render the pinned exact certificates; the small scale difference is quantified above.

The improvement is in the contact return.

A physical splice changes which chain closes; the remaining constraints determine the angle.

1 Start from an actual contact skeleton

The underlying 70-square construction is due to **Ryan Xu**. Splicing physical pieces of that construction gives a complete 153-square assembly. The new geometry keeps whole unit squares and the specified contact relations.

2 Replace the forcing cycle

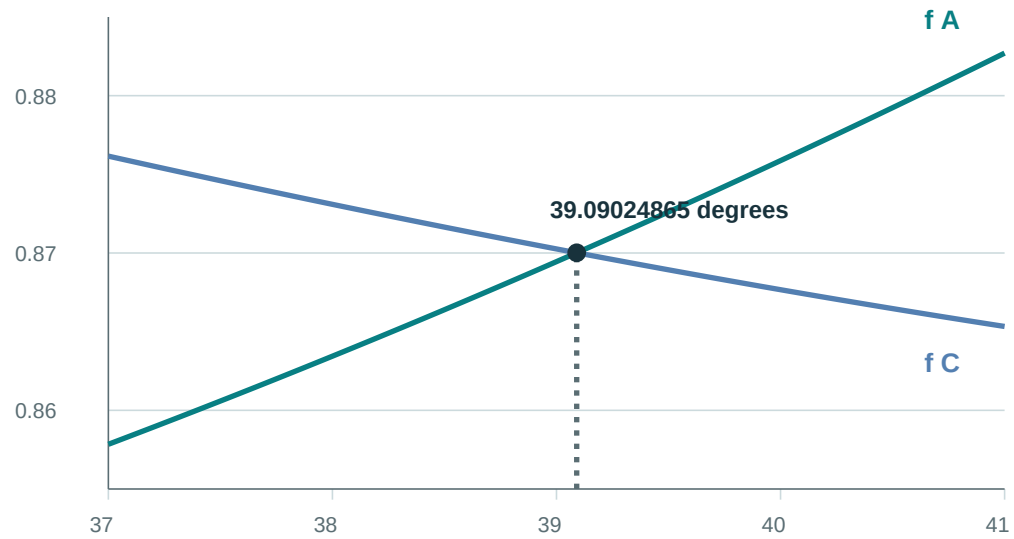
The splice removes the original long forcing return. A shorter half-return now ends at an upright square, which supplies its cap. This changes the controlling boundary equation while preserving the other, short return.

3 Solve the two surviving returns

Write the enclosing side as **L = 12 + f**. The retained short chain requires **f** ≥ $\mathbf{f_A}$; the newly capped chain requires **f** ≥ $\mathbf{f_C}$. Their crossing selects the candidate shown here. The angle is derived from these contact equations.

Short return: $f_A = (2 - \cos(\theta)) / (\cos(\theta) + \sin(\theta))$

TWO REMAINING CONTACT BOUNDS



Bounds within this contact family; the crossing is not a proof of global optimality.

With **u = tan(theta / 2)**, the crossing reduces to:

$$u^7 - 9u^6 + 3u^5 - u^4 - u^3 + u^2 - 3u + 1 = 0$$

u = 0.355005149036226...
L before the certification guard = 12.870013850404464...

INDEPENDENT EXACT VERIFICATION

153 unit squares. 11,628 pairs. All pass.
Rational support tests check every pair without a bounding-box shortcut. A separate polygon check confirms unit edges, containment and zero overlap area.

The certificate uses exact rational centers and rational half-angle parameters. Its certified side is **12.870013850407044...**; the headline rounds upward. A center-and-container dilation of **$1 + 2 \times 10^{-13}$** supplies the guard; square sides stay exactly one.

Construction credit: **PingYou Ltd**. Underlying N=70 source: **Ryan Xu**. The companion package includes the pinned public comparison, exact witness, return derivation and independent checks.